

SLIDE 1 — What is Array Indexing?

Definition:

An **index** is a number used to identify the position of an element in an array.

Simple idea

Array → **Index** → **Element**

Example:

```
int marks[5] = {80, 90, 75, 85, 95};  
Index:  0   1   2   3   4  
       [80] [90] [75] [85] [95]
```

Explanation:

The first element is at **index 0**.

 **Remember: Index = Position**

SLIDE 2 — How Does Indexing Work?

Structure

```
arrayName[index];
```

Example

```
int numbers[5] = {10, 20, 30, 40, 50};  
  
printf("%d", numbers[2]);
```

Explanation:

```
numbers[2]  
  ↓  
Index 2  
  ↓  
Value 30
```

Output:

30

★ Remember

- First element → 0
 - Second element → 1
 - Last element → size - 1
-

SLIDE 3 — Accessing an Element

An array element is accessed using its **index**.

Example

```
int marks[5] = {85, 90, 78, 92, 88};
```

```
printf("%d", marks[2]);
```

Explanation:

`marks[2]` accesses the **third element**.

Output:

78

Quick View

```
Index: 0  1  2  3  4
       [85] [90] [78] [92] [88]
           ↑
        marks[2]
```

🧠 Remember: `array[index]` → Access an element

SLIDE 4 — Changing an Element

Array elements can be changed using their index.

Example

```
int marks[5] = {80, 90, 75, 85, 95};
```

```
marks[2] = 100;
```

Before

```
[80] [90] [75] [85] [95]
```

↑
index 2

After

```
[80] [90] [100] [85] [95]
```

↑
changed

⚠ Important

For:

```
int marks[5];
```

Valid indexes are:

0, 1, 2, 3, 4

🧠 **Basic Pattern:**

INDEX → FIND → READ / CHANGE

SLIDE 5 — RECAP

Access Elements Using Index! 🔍

Flow

ARRAY → INDEX → POSITION → ELEMENT

🎯 5 Things to Remember

1. Index identifies an element
2. Index starts from 0
3. Use `array[index]` to access
4. Last index = size - 1
5. Elements can be read or changed

Final Example

```
int numbers[5] = {10, 20, 30, 40, 50};
```

```
printf("%d", numbers[3]);
```

3 → 40

Output:

40

Memory Trick

Index → POSITION 

`array[index]` → ACCESS 

Starts at 

QUESTIONS

1. What is the first index of an array in C?

```
int numbers[5] = {10, 20, 30, 40, 50};
```

 **Hint:** C starts counting from zero.

 **Glossary:** Index → Position used to access an element.

✓ Answer: 0

2. Access an Element

Complete the code to print the third element of the array.

```
int numbers[5] = {10, 20, 30, 40, 50};
```

```
printf("%d", numbers[____]);
```

💡 Hint: Array indexing starts from 0.

Answer: 2

3. What is the output?

```
int numbers[5] = {10, 20, 30, 40, 50};
```

```
printf("%d", numbers[2]);
```

💡 Hint: Start counting from index 0.

📖 Glossary: Access → Get an element from an array.

✓ Answer: 30

4. Arrange the steps:

- A. Get the element
- B. Choose an index
- C. C finds the position
- D. Access using array[index]

💡 Hint: Choose → Access → Find → Get.

📖 Glossary: Position → Location of an element.

✓ Answer: B → D → C → A

5. Read and Change an Element

Complete the code to change the second element from 90 to 100.

```
int marks[5] = {80, 90, 75, 85, 95};
```

```
marks[____] = 100;
```

```
printf("%d", marks[1]);
```

💡 Hint: The second element has index 1.

Answer: **1**